

Per-transfer sunset hardening: the endpoint joint is no longer thin (S = x1.129 -> x2.994 realized)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The B5 T6-conditional hypothesis H-ENDPOINT-THINNESS-ACCEPTED exists because the certified endpoint joint saturates at the sunset cap $S = \times 1.13$. This note attacks that cap with the methodology that certified the quartic channel: replace a sup-bound kernel by its realized per-transfer evaluation. **Scope:** the sunset kernel factor of the endpoint bookkeeping, at the endpoint dressing, on the admissible chord set. **Not claimed:** any gate or tier change; any new analytic theorem (this is an EXECUTED evaluation of the existing bound's kernel, certificate-candidate grade).

2. Pre-registered gates (set before computation)

G1: $\text{shape}_{\max} = \max_{t \in \text{chords}} J_{\text{eff}}(t)/J_0 \geq 1 \Rightarrow \text{hardening FAILS (honest negative)}$. G2: $S_{\text{realized}} \leq 1.2 \Rightarrow \text{immaterial}$. G3: $\text{success} \Rightarrow \text{report } S_{\text{realized}}$ and the recomputed joints at $\rho_{\text{lat}} = 6.55$ and 12.6. **Outcome: G3.**

3. The replacement and why it is exact

The anchor bound (scscope-mendpoint-evaluation v1.1) is $b = \frac{1}{2} u_{\text{eff}}^2(2I) 6 J_0 M$ with $J_0 = \sup_t J(t)$ at the 4×10^{-4} dressing: the 3-propagator sunset kernel $(J * G)(t)$ (bubble $\times$ propagator) bounded by $\sup J \cdot \int G$. The realized replacement evaluates the convolution instead of bounding it. **Derivation compatibility (explicit):** the existing bookkeeping multiplies the TOTAL transfer weight by ONE kernel constant; ANY constant that upper-bounds the kernel on every realized transfer is admissible there, and $\max_{t \in \text{chords}} J_{\text{eff}}(t)$ is such a constant (machine-asserted $J_{\text{eff}}(t) \leq J_0$ pointwise and every admissible transfer lies in the chord set) – so the replacement is a strictly valid sharpening of the same inequality, not a new bookkeeping. The kernel is:

$$J_{\text{eff}}(t) = \frac{N(t)}{N_0}, \quad N(t) = \iint \frac{k^2}{D(k)} J(\sqrt{k^2 + t^2 + 2kt\mu}) d\mu dk,$$

$$N_0 = \iint \frac{k^2}{D(k)} d\mu dk, \quad D(k) = \hat{r}_{\text{end}} + C(k^2 - q_0^2)^2,$$

with ALL propagators at the endpoint dressing $\hat{r}_{\text{end}} = 0.33675$. The absolute normalisation cancels in the ratio: no $(2\pi)^3$ convention enters (the factor-2 error class is structurally excluded), and $J_{\text{eff}}(t) \leq J_0$ pointwise (machine assert). Admissible transfers obey $|t| \geq t_{\min} = 2q_0 \sin(\theta_{\min}/2) = 0.420$, so the chord set $[t_{\min}, 2q_0]$ covers every realized transfer.

4. Results (scscope_sunset_pertransfer.py, 8/8 asserts PASS)

Calibration: the anchor is REPRODUCED, $S_{\text{anchor}} = \times 1.129$ ($J_0 = 0.2896$, $M_{\text{END}} = 0.104953$). Two-way pin: $J_{\text{eff}}(t \rightarrow 0)$ by the convolution and by the degenerate 1D integral agree to 10^{-5} . Grid-doubling drift on the worst chord: 0.00%.

$$\text{shape}_{\text{max}} = \frac{\max_t J_{\text{eff}}(t)}{J_0} = \frac{0.1093}{0.2896} = 0.377 \quad (\text{at } t = t_{\text{min}} = 0.420),$$

$$S_{\text{realized}} = \frac{1.129}{0.377} = \times 2.994.$$

Recomputed joints (certified $R_{\text{max}} = 0.385$ from the quartic certificate run artefact):

$$\text{joint}(\rho=6.55) : \times 1.040 \rightarrow \times 2.023, \quad \text{joint}(\rho=12.6) : \times 1.082 \rightarrow \times 2.396,$$

$$\text{cap}(\rho \rightarrow \infty) : \times 1.13 \rightarrow \times 2.994.$$

The doubly-conservative structure of the anchor (sup over transfer; loosest dressing) accounts for the factor: the loop average of J at $|t| \geq 0.420$ sits far below the $t \rightarrow 0$ sup, exactly as the quartic’s Young-vs-realized gap ($1.019 \rightarrow 0.385$) did.

5. Consequence for the B5 T7 path (no flip enacted)

If the operator accepts this evaluation as the sunset accounting, the endpoint closure is COMFORTABLE ($\times 2.0$ – $\times 2.4$), and H-ENDPOINT-THINNESS-ACCEPTED – one of the two $\mathcal{H}_{\text{B5}}^{\text{T6}}$ hypotheses – loses its load: its removal would leave H-NONLATTICE-REMAINDER-EXCLUDED (T-030) as the single remaining B5 T7 blocker. This note enacts NOTHING: the acceptance, the GATES.md update, the W_SC re-certification with the hardened S, and any SC-SCOPE package re-issue are operator decisions upon review.

5b. Mixed-dressing justification (operator-required for the final stamp)

Monotonicity argument. Both kernel factors move the SAME way with the dressing: $J(t; r)$ decreases in r (heavier mass, smaller bubble – the direction already certified by the endpoint table $J(\hat{r}(I)) \leq J_0$), and the loop weight $1/D(k; r)$ decreases in r . Hence over ALL assignments of dressings to the sunset’s propagators, the LIGHTEST (anchor, 4×10^{-4}) assignment maximises J_{eff} : it is the adversarial corner, and matched endpoint dressing is the physically correct (per-pattern self-consistent) choice, used non-load-bearingly.

Machine worst-case (script v1.1.0, asserts 9–10). All four (J -grid, D) dressing assignments evaluated on the chords: worst J_{eff} per variant = $0.1093/0.1211/0.1110/0.1230$ (end-end / anchor-end / end-anchor / anchor-anchor), confirming the monotone ordering. At the adversarial corner: $\text{shape}_{\text{worst}} = 0.4247$, $S_{\text{worst}} = \times 2.659 > 2$, $\text{joint}(\rho=6.55) = \times 1.886$ — the hardening and the gate-removal claim SURVIVE the worst dressing assignment.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “The weighted-average replacement may be doing bookkeeping work the bound’s derivation does not permit (the sum over transfers might need the kernel at EACH transfer, not a sup).” ADDRESSED: the original bound already uses ONE kernel value (the sup) times the total weight; replacing the global sup by the sup over the ADMISSIBLE chord set is a strictly valid intermediate ($J_{\text{eff}}(t) \leq \max_{\text{chords}} J_{\text{eff}} \leq J_0$ pointwise on every realized transfer); no per-transfer redistribution is claimed.

- (β) “The dressing choice (all propagators at $\hat{r}_{\text{end}}$) could be optimistic.” RESOLVED in v1.2 (operator-required): Sec. 5b gives the monotonicity argument AND the machine worst-case – even the adversarial (anchor, anchor) assignment leaves $S_{\text{worst}} = \times 2.659 > 2$ and joint = $\times 1.886$; the matched endpoint choice is not load-bearing.
- (γ) “ J_{eff} peaks at t_{min} – a competitor just above the admissibility floor gets the weakest hardening; is the floor itself certified?” DISMISSED: t_{min} comes from $\theta_{\text{min}}(\hat{r})$, the SAME admissibility input the whole B5 chain uses (machine-asserted here); at t_{min} the hardening is already $\times 2.65$ relative to J_0 , and the gate G3 threshold was pre-registered well below the observed value.

Quantitative sanity check (mandatory). Direction: a smaller kernel can only RAISE the sunset margin – selection-favorable by construction, and the mechanism (loop average < sup) is fixed by $J(t)$ monotonicity on the chord range, not tuning. Limit case: at $t \rightarrow 0$ the two-way pin agrees to 10^{-5} ; at the anchor inputs the script REPRODUCES $\times 1.129$ before any replacement. Magnitude: $\text{shape}_{\text{max}} = 0.377$ is comparable to the quartic’s realized/Young ratio ($0.385/1.019 = 0.378$) – the same kernel geometry drives both, an independent cross-consistency.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / SC-SCOPE sunset per-transfer hardening (B5 T7 path)
Precise statement:	EXECUTED (8/8): replacing the single-J0 sup anchor by the realized D-weighted loop average $J_{\text{eff}}(t)$ on the admissible chords (all propagators at the endpoint dressing) gives $\text{shape}_{\text{max}} = 0.377$, $S = \times 1.129 \rightarrow$ $S_{\text{realized}} = \times 2.994$; endpoint joint $\times 1.040$ $\rightarrow \times 2.023$ ($\rho=6.55$), $\times 1.082 \rightarrow \times 2.396$ ($\rho=12.6$); saturation cap $\times 1.13 \rightarrow \times 2.994$. Gate G3-SUCCESS (pre-registered).
Scope:	the sunset kernel factor at the endpoint, on the admissible chord set. Certificate- candidate (EXECUTED evaluation); no new analytic theorem; no flip.
Evidence grade:	EXECUTED certificate-candidate (T4-grade evidence for the operator decision).
Reproduction command:	python codes/vacuum/scscope_sunset_pertransfer.py
Falsifier:	a realized admissible transfer with $J_{\text{eff}}(t) > \max_{\text{chords}} J_{\text{eff}}(\text{chord-set}$ escape); anchor non-reproduction; grid drift > 1%.
Tier before / after:	PUBLISHED-BUNDLE CONFIRMED (operator final ratification 2026-06-12; clean-run 10/10); H-ENDPOINT-THINNESS-ACCEPTED REMOVED (FINAL); B5 T6-conditional with the single hypothesis H-NONLATTICE-REMAINDER-EXCLUDED; bundle SC-SCOPE-SunsetHardened-T6-260612.
No-overclaim statement:	hardening of the KERNEL FACTOR only, within the existing bookkeeping (derivation- compatibility stated explicitly, Sec. 3); the mixed-dressing audit is DELIVERED (Sec. 5b, worst case $S = \times 2.659$); W_{SC} re-certification with the hardened S is not done here.
Next required action:	none for this result (PUBLISHED). Mainline: T-030 = the single remaining B5 T7 blocker.